# Assignment 23

## Dataset

| Patient | Age | Chol | BP | Target |
|---|---|---|---|---|
| P1 | 55 | 250 | 140 | 1 |
| P2 | 60 | 240 | 150 | 1 |
| P3 | 45 | 230 | 130 | 1 |
| P4 | 50 | 220 | 120 | 0 |
| P5 | 48 | 210 | 125 | 0 |
| P6 | 52 | 215 | 135 | 0 |

$P(1) = 3/6 = 0.5$
$P(0) = 3/6 = 0.5$

**Mean ($\mu$)**

for 1

$\mu\_age = (55+60+45)/3 = 53.3$

for 0

$\mu\_age = (50+48+52)/3 = 50$

**Deviation from Mean**

| Value (x) | Squares |
|---|---|
| $55 - 53.3 = 1.7 \Rightarrow$ | 2.89 |
| $60 - 53.3 = 6.7 \Rightarrow$ | 44.89 |
| $45 - 53.3 = -8.3 \Rightarrow$ | 68.89 |

Variance

$(2.89 + 44.89 + 68.89)/3 = 116.67/3 = 38.89$

Standard deviation

$\text{sqrt}(38.89) = 6.2$

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Disease (1)

Chol = 250, 240, 230

Mean = (250 + 240 + 230) / 3 = 720/3 = 240

Deviation

| (Value − Mean) | Square |
|---|---|
| 250 − 240 = 10 | 100 |
| 240 − 240 = 0 | 0 |
| 230 − 240 = −10 | 100 |

Variance

(100 + 0 + 100) / 3 = 200/3 = 66.67

Standard deviation

sqrt (66.67) = 8.2

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Use of gaussian ( Same Dataset )

Criteria = Age = 54

Chol = 245

BP = 138

Class Probabilities

Total = 6

(1) = 3        P(1) = 3/6 = 0.5

(0) = 3        P(0) = 3/6 = 0.5

(μ)

for (1)

μ_age = (55 + 60 + 45) / 3 = 160/3 = 53.3

for (0)

μ_age = (50 + 48 + 52) / 3 = 50

Age = 53.3

Chol = 250, 240, 230

Mean = (250 + 240 + 230) | 3
= 240

BP = 140

No disease (0):
Age = 50
Chol = 215
BP = 126.67

SD:

| for (1) | | for (0) | |
|---|---|---|---|
| Age | = 6.2 | Age | = 2 |
| Chol | = 8.2 | Chol | = 4.1 |
| BP | = 8.2 | BP | = 4.2 |

gaussian
Criteria Age = 54    Chol = 245    BP = 138

for disease (1)
Age: sqrt (54 - 53.3) | (2 X 6.2 X 6.2) = 0.98

Chol 245 = 0.83
BP 138 = 0.97

Disease (1) → 0.5 X 0.98 X 0.83 X 0.97
= 0.395

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for disease(0)

Age = 54 P = 0.13
Chol = 245 P = 0.01
BP = 138 P = 0.2

Disease (0) = 0.5 × 0.13 × 0.01 × 0.2 = 0.00013

Disease (1) → 0.395
Disease (0) → 0.00013